

- 4 (a) Gas molecules move randomly within the cylinder. The molecules collide with the wall of the cylinder and piston. Upon collisions with the piston, the molecules will undergo a change of their momentum. Since the system (comprising the gas molecules and piston) is closed and no external force acts on it, the law of conservation of linear momentum applies. By Newton's 2nd law, the rate of change in momentum gives the average force that the piston exerts on the molecules to cause this rebound. By Newton 3rd Law, the gas molecules will exert an equal magnitude average force on piston.
- (b) (i) The higher temperature will cause the gas molecules' root mean square speed to increase. The average kinetic energy of the molecules increase. There is a greater change in momentum when the gas molecules collide with the piston. Collision frequency with the piston increases. A greater average force is exerted on the piston. The piston is moved outwards. The volume of the gas increases. As volume of gas increases, there is a greater distance for gas molecules to travel in between collisions. Collision frequency with the piston decreases. Force on the piston remains constant as the rate of change of momentum is less. Hence pressure remains constant as volume increases.

(ii)

$$pV = nRT$$

since pressure is kept constant and no change in amount of gas, $V \propto T$

$$\text{For } 303 \text{ K, } V_1 \propto T_1 \dots(1)$$

$$\text{For } 363 \text{ K, } V_2 \propto T_2 \dots(2)$$

$$\frac{(2)}{(1)}: V_2 = \frac{363}{303} \times 1.6 \times 10^{-3} = 1.9 \times 10^{-3} \text{ m}^3$$

(iii) Work done by gas = $(1.9 - 1.6) \times 10^{-3} \times 1.1 \times 10^5 = 33 \text{ J}$

- (c) There is a change in internal energy of the gas. This change in the internal energy is equal to the sum of heat energy supplied to gas and work done on the gas. Since the gas volume increases, work is done by the gas. Hence, work done on the gas is negative.

In mathematical equation, the first law is given as $\Delta U = Q + W$

- 5 (a) The induced e m f is given by the rate of change of flux linkage